jupyter Experiment4 Last Checkpoint: 22 days ago

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Code

JupyterLab Python (ml env)

[1]:

```
a = 20
b = 6

addition = a + b
subtraction = a - b
multiplication = a * b
division = a / b
floor_division = a // b
modulus = a % b
exponent = a ** b

print(f"Numbers: a = {a}, b = {b}")
print("-" * 30)
print(f"Addition (a + b):         {addition}")
print(f"Subtraction (a - b):      {subtraction}")
print(f"Multiplication (a * b):   {multiplication}")
print(f"Division (a / b):         {division}")
print(f"Floor Division (a // b):   {floor_division}")
print(f"Modulus / Remainder (a % b): {modulus}")
print(f"Exponent (a ** b):        {exponent}")
```

```
Numbers: a = 20, b = 6
------------------------------
Addition (a + b):         26
Subtraction (a - b):      14
Multiplication (a * b):   120
Division (a / b):         3.3333333333333335
Floor Division (a // b):   3
Modulus / Remainder (a % b): 2
Exponent (a ** b):        64000000
```

[3]:

```
import math
import numpy as np
import pandas as pd

data = [12, 15, 18, 20, 22, 25, 28, 30, 35, 40]

print("--- 1. Using NumPy ---")
mean_np = np.mean(data)
median_np = np.median(data)
std_population = np.std(data)          # ddof=0 (population standard deviation)
std_sample = np.std(data, ddof=1)      # ddof=1 (sample standard deviation)
var_sample = np.var(data, ddof=1)      # sample variance
data_range = np.ptp(data)              # peak-to-peak (max - min)
q25, q75 = np.percentile(data, [25, 75])
iqr = q75 - q25                        # Interquartile Range

print(f"Mean:                      {mean_np:.2f}")
print(f"Median:                    {median_np:.2f}")
print(f"Sample Std Dev (s):        {std_sample:.2f}")
print(f"Population Std Dev (sigma): {std_population:.2f}")
print(f"Sample Variance (s^2):     {var_sample:.2f}")
print(f"Range (Max - Min):         {data_range}")
print(f"IQR (Q3 - Q1):             {iqr:.2f}")


print("\n--- 2. Using Pandas (Full Summary) ---")
series = pd.Series(data)

summary = series.describe()
print(summary)


print("\n--- 3. Differences / Residuals (xi - mean) ---")

deviations = [x - mean_np for x in data]
for original, diff in zip(data, deviations):
    print(f"Value: {original:2d} | Deviation from mean: {diff:+6.2f}")
```

```
--- 1. Using NumPy ---
Mean:                      24.50
Median:                    23.50
Sample Std Dev (s):        8.87
```

```
    print(f"Value: {original:2d} | Deviation from mean: {diff:+6.2f}")
```

```
--- 1. Using NumPy ---
Mean:                       24.50
Median:                     23.50
Sample Std Dev (s):         8.87
Population Std Dev (sigma): 8.42
Sample Variance (s^2):      78.72
Range (Max - Min):          28
IQR (Q3 - Q1):              11.00

--- 2. Using Pandas (Full Summary) ---
count    10.000000
mean     24.500000
std       8.872554
min      12.000000
25%      18.500000
50%      23.500000
75%      29.500000
max      40.000000
dtype: float64

--- 3. Differences / Residuals (xi - mean) ---
Value: 12 | Deviation from mean: -12.50
Value: 15 | Deviation from mean:  -9.50
Value: 18 | Deviation from mean:  -6.50
Value: 20 | Deviation from mean:  -4.50
Value: 22 | Deviation from mean:  -2.50
Value: 25 | Deviation from mean:  +0.50
Value: 28 | Deviation from mean:  +3.50
Value: 30 | Deviation from mean:  +5.50
Value: 35 | Deviation from mean: +10.50
Value: 40 | Deviation from mean: +15.50
```